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FACULTY OF INFORMATION TECHNOLOGY

Library Management System

Project Information

Project Name	Library Mangement System
Course Title	Introduction To Computing - LAB
Submission Date	17-June-2025
Faculty	Faculty Of Information Technology



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Overview

This **C++** program is a comprehensive **Library Management System** that allows librarians to manage books, members, and book transactions efficiently. It provides a console-based interface with various functionalities for library operations.

Key Features

1. User-Friendly Menu System

- Clear numbered options (**1-14**) for all major functions
- Input validation to prevent errors
- Visual and audio feedback (**beeps**) for user actions

2. Book Management

- Add new books with complete details (**ID, title, author, category, price**)
- Update existing book information
- Remove books from the system
- View all books with their current status
- Search books by title, author, or category

3. Member Management

- Register new members (**students or teachers**)
- Update member details
- Different member types with different privileges
- Unique member IDs to prevent duplicates

4. Transaction System

- Issue books to members with type-based limits (**students: 2 books, teachers: 3 books**)
- Return books and update status
- Track which books are issued to which members



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5. Data Validation

- Strict input validation for all fields
- Proper data type checking
- Format requirements for names, titles, etc.
- Prevention of duplicate member IDs

6. Reporting

- View total number of books issued
- Detailed book status information (available/issued)
- Search functionality with comprehensive results

Functionality

The system maintains two main sets of data:

- **Books:** Stored in arrays with details like title, author, category, price, and issue status
- **Members:** Stored in arrays with name, ID, type (student/teacher), and books issued count

The program uses a while loop to continuously display the menu and process user choices until the exit option is selected. Each menu option is implemented with proper error handling and user feedback.

Technical Details

- Uses basic C++ constructs (arrays, loops, conditionals)
- Implements input/output operations with validation
- Includes Windows-specific features for sound (Beep) and console control
- Maintains data in memory during program execution
- Has maximum limits for books (100) and members (50)



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Pescudo Code

BEGIN

// Constants

MAX_BOOKS = 100

MAX_MEMBERS = 50

// Arrays for storing data

DECLARE bookTitle[MAX_BOOKS], bookAuthor[MAX_BOOKS], bookCategory[MAX_BOOKS] **AS**

STRING

DECLARE bookPrice[MAX_BOOKS] **AS** FLOAT

DECLARE bookID[MAX_BOOKS], issuedBooks[MAX_BOOKS] **AS** INTEGER

DECLARE memberName[MAX_MEMBERS] **AS** STRING

DECLARE memberID[MAX_MEMBERS], booksIssuedCount[MAX_MEMBERS] **AS** INTEGER

DECLARE memberType[MAX_MEMBERS] **AS** CHAR

// Counters

totalBooks = 0

totalMembers = 0

// Main program loop

WHILE TRUE **DO**

DISPLAY_MENU()

GET userChoice **FROM** INPUT

IF userChoice **is** invalid **THEN**

SHOW error message

CONTINUE

END IF

CASE userChoice **OF**

 1: // Issue Book

GET memberID **FROM** USER

FIND member



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IF member not found THEN

 SHOW error

 CONTINUE

END IF

CHECK issue limit based on member type

IF limit reached THEN

 SHOW error

 CONTINUE

END IF

GET bookID FROM USER

FIND book

IF book not found THEN

 SHOW error

ELSE IF book is available THEN

 ISSUE book to member

 INCREMENT member's issued count

 SHOW success

ELSE

 SHOW "Book already issued"

END IF

2: // Return Book

GET bookID FROM USER

FIND book

IF book not found THEN

 SHOW error

ELSE IF book is issued THEN

 DECREMENT member's issued count

 MARK book as available

 SHOW success

ELSE

 SHOW "Book was not issued"

END IF



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3: // Add/Buy Book

```
IF library is full THEN
    SHOW error
ELSE
    GET book details FROM USER
    VALIDATE all inputs
    ADD new book
    INCREMENT totalBooks
    SHOW success
END IF
```

4: // Register Member

```
IF members limit reached THEN
    SHOW error
ELSE
    GET member details FROM USER
    VALIDATE all inputs
    ADD new member
    INCREMENT totalMembers
    SHOW success
END IF
```

5: // Update Member

```
GET memberID FROM USER
FIND member
IF found THEN
    GET new details FROM USER
    VALIDATE inputs
    UPDATE member
    SHOW success
ELSE
    SHOW "Member not found"
END IF
```



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6: // Total Books Issued

COUNT issued books

DISPLAY count

7: // Update Book

GET bookID FROM USER

FIND book

IF found THEN

 GET new details FROM USER

 VALIDATE inputs

 UPDATE book

 SHOW success

ELSE

 SHOW "Book not found"

END IF

8: // Remove Book

GET bookID FROM USER

FIND book

IF found THEN

 IF book is issued THEN

 DECREMENT member's issued count

 END IF

 REMOVE book

 DECREMENT totalBooks

 SHOW success

ELSE

 SHOW "Book not found"

END IF

10: // View All Books

DISPLAY all books with details and status



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11: // Search by Title

GET title FROM USER

SEARCH and DISPLAY matching books

12: // Search by Author

GET author FROM USER

SEARCH and DISPLAY matching books

13: // Search by Category

GET category FROM USER

SEARCH and DISPLAY matching books

14: // Exit

CONFIRM exit

IF confirmed THEN

BREAK loop

END IF

END CASE

WAIT for user to press Enter

END WHILE

END

Code on VS STUDIO :

```
#include <iostream>
#include <windows.h> // For color and sound
#include <string>
using namespace std;

int main() {
    const int maxBooks = 100;
```



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```
const int maxMembers = 50;
string bookTitle[maxBooks];
string bookAuthor[maxBooks];
string bookCategory[maxBooks];
float bookPrice[maxBooks];
int bookID[maxBooks];
int totalBooks = 0;
string memberName[maxMembers];
int memberID[maxMembers];
char memberType[maxMembers]; // 'S' for student, 'T' for teacher
int booksIssuedCount[maxMembers] = { 0 }; // Track books issued per
member

int totalMembers = 0;
int issuedBooks[maxBooks] = { 0 }; // 0 = Available, 1 = Issued (stores
member ID)

while (true) {
    system("cls");
    cout << "\n\n
===== LIBRARY MANAGEMENT SYSTEM =====\n\n" << endl;
    cout << "1. Issue Book" << endl;
    cout << "2. Return Book" << endl;
    cout << "3. Buy Book" << endl;
    cout << "4. Register Member" << endl;
    cout << "5. Update Member Details" << endl;
    cout << "6. Total Books Issued" << endl;
    cout << "7. Add Book" << endl;
    cout << "8. Update Book" << endl;
    cout << "9. Remove Book" << endl;
    cout << "10. View All Books" << endl;
    cout << "11. Search Book by Title" << endl;
    cout << "12. Search Book by Author" << endl;
    cout << "13. Search Book by Category" << endl;
    cout << "14. Exit\n" << endl;
```



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```
cout << "Enter your choice (1-14): ";

int choice;

if (!(cin >> choice) || choice < 1 || choice > 14) {
    cin.clear(); // Clear error state
    cin.ignore(10000, '\n'); // Discard invalid input
    cout << "Invalid input! Please enter a number (1-14)." <<
endl;

    Beep(500, 500); // Error sound
    Sleep(1000); // 1 sec delay
    continue;
}

cin.ignore(10000, '\n'); // Clear the newline after number input


if (choice == 1) { // Issue Book
    cout << "Enter Member ID: ";
    int memberId;
    if (!(cin >> memberId)) {
        cout << "Invalid Member ID! Please enter a numeric
value." << endl;

        Beep(500, 500);
        cin.clear();
        cin.ignore(10000, '\n');
        Sleep(1000);
        continue;
    }
    cin.ignore(10000, '\n');

    // Find member
    int memberIndex = -1;
    char type = ' ';
    for (int i = 0; i < totalMembers; i++) {
        if (memberID[i] == memberId) {
            memberIndex = i;
            type = memberType[i];
        }
    }
}
```



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break;

}

}

```
if (memberIndex == -1) {
```

```
    cout << "Member not found!" << endl;
```

```
    Beep(500, 500);
```

```
    cout << "\nPress Enter to continue...";
```

```
    cin.ignore(10000, '\n');
```

```
    continue;
```

```
}
```

```
// Check issuing limit
```

```
if ((type == 'S' && booksIssuedCount[memberIndex] >= 2) ||
```

```
    (type == 'T' && booksIssuedCount[memberIndex] >= 3)) {
```

```
    cout << "Issue limit reached! ";
```

```
    if (type == 'S') cout << "Students can issue max 2
```

```
books." << endl;
```

```
    else cout << "Teachers can issue max 3 books." << endl;
```

```
    Beep(500, 500);
```

```
    cout << "\nPress Enter to continue...";
```

```
    cin.ignore(10000, '\n');
```

```
    continue;
```

```
}
```

```
cout << "Enter Book ID to issue: ";
```

```
int bookId;
```

```
if (!(cin >> bookId)) {
```

```
    cout << "Invalid Book ID! Please enter a numeric value."
```

```
<< endl;
```

```
    Beep(500, 500);
```

```
    cin.clear();
```

```
    cin.ignore(10000, '\n');
```

```
    Sleep(1000);
```



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continue;

}

cin.ignore(10000, '\n');

bool bookFound = false;

for (int i = 0; i < totalBooks; i++) {

if (bookID[i] == bookId) {

bookFound = true;

if (issuedBooks[i] == 0) {

issuedBooks[i] = memberId; // Store member ID

who issued the book

booksIssuedCount[memberIndex]++;

cout << "Book issued successfully to "

<< (type == 'S' ? "Student" : "Teacher")

<< " " << memberName[memberIndex] << "!"

<< endl;

Beep(1000, 200);

}

else {

cout << "Book is already issued!" << endl;

Beep(500, 500);

}

break;

}

}

if (!bookFound) {

cout << "Book not found!" << endl;

Beep(500, 500);

}

cout << "\nPress Enter to continue...";

cin.ignore(10000, '\n');

}

else if (choice == 2) { // Return Book



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```
cout << "Enter Book ID to return: ";

int bookId;

if (!(cin >> bookId)) {
    cout << "Invalid Book ID! Please enter a numeric value."
<< endl;

    Beep(500, 500);
    cin.clear();
    cin.ignore(10000, '\n');
    Sleep(1000);
    continue;
}

cin.ignore(10000, '\n');

bool bookFound = false;
for (int i = 0; i < totalBooks; i++) {
    if (bookID[i] == bookId) {
        bookFound = true;
        if (issuedBooks[i] != 0) {
            // Find member to decrement their issued count
            int memberId = issuedBooks[i];
            for (int j = 0; j < totalMembers; j++) {
                if (memberID[j] == memberId) {
                    booksIssuedCount[j]--;
                    break;
                }
            }
            issuedBooks[i] = 0;
            cout << "Book returned successfully!" << endl;
            Beep(1000, 200);
        }
    }
    else {
        cout << "Book was not issued!" << endl;
        Beep(500, 500);
    }
}
```



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```
break;
```

```
}
```

```
}
```

```
if (!bookFound) {
```

```
    cout << "Book not found!" << endl;
```

```
    Beep(500, 500);
```

```
}
```

```
cout << "\nPress Enter to continue...";
```

```
cin.ignore(10000, '\n');
```

```
}
```

```
else if (choice == 3 || choice == 7) { // Buy Book or Add Book
```

```
    if (totalBooks < maxBooks) {
```

```
        cout << "Enter Book ID: ";
```

```
        int id;
```

```
        if (!(cin >> id)) {
```

```
            cout << "Invalid Book ID! Please enter a numeric
```

```
value." << endl;
```

```
            Beep(500, 500);
```

```
            cin.clear();
```

```
            cin.ignore(10000, '\n');
```

```
            Sleep(1000);
```

```
            continue;
```

```
        }
```

```
        cin.ignore(10000, '\n');
```

```
        bookID[totalBooks] = id;
```

```
        string input;
```

```
        bool valid = false;
```

```
        // Title validation
```

```
        do {
```

```
            cout << "Enter Title (letters, numbers, and spaces
```

```
only): ";
```



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```
getline(cin, input);
valid = true;
for (char c : input) {
    if (!((c >= 'a' && c <= 'z') || (c >= 'A' && c
<= 'Z') ||

(c >= '0' && c <= '9') || c == ' ')) {
        valid = false;
        cout << "Invalid Title! Only letters,
numbers and spaces allowed." << endl;
        Beep(500, 500);
        Sleep(500);
        break;
    }
}
} while (!valid);
bookTitle[totalBooks] = input;

// Author validation (letters and spaces only)
do {
    cout << "Enter Author (letters and spaces only): ";
    getline(cin, input);
    valid = true;
    for (char c : input) {
        if (!((c >= 'a' && c <= 'z') || (c >= 'A' && c
<= 'Z') || c == ' ')) {
            valid = false;
            cout << "Invalid Author! Only letters and
spaces allowed." << endl;

            Beep(500, 500);
            Sleep(500);
            break;
        }
    }
} while (!valid);
```




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```
bookAuthor[totalBooks] = input;
```

```
// Category validation
```

```
do {
```

```
    cout << "Enter Category (letters, numbers, spaces,  
and underscores only): ";
```

```
    getline(cin, input);
```

```
    valid = true;
```

```
    for (char c : input) {
```

```
        if (!((c >= 'a' && c <= 'z') || (c >= 'A' && c  
<= 'Z') ||
```

```
            (c >= '0' && c <= '9') || c == ' ' || c  
== '_')) {
```

```
            valid = false;
```

```
            cout << "Invalid Category! Only letters,  
numbers, spaces and underscores allowed." << endl;
```

```
            Beep(500, 500);
```

```
            Sleep(500);
```

```
            break;
```

```
        }
```

```
    }
```

```
} while (!valid);
```

```
bookCategory[totalBooks] = input;
```

```
// Price validation
```

```
float price;
```

```
while (true) {
```

```
    cout << "Enter Price: $";
```

```
    if (!(cin >> price)) {
```

```
        cout << "Invalid Price! Please enter a numeric  
value." << endl;
```

```
        Beep(500, 500);
```

```
        cin.clear();
```

```
        cin.ignore(10000, '\n');
```



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Sleep(500);

}

else {

cin.ignore(10000, '\n');

break;

}

}

bookPrice[totalBooks] = price;

issuedBooks[totalBooks] = 0;

totalBooks++;

cout << "Book added successfully!" << endl;

Beep(1000, 200);

}

else {

cout << "Library is full! Cannot add more books." <<

endl;

Beep(500, 500);

}

cout << "\nPress Enter to continue...";

cin.ignore(10000, '\n');

}

else if (choice == 4) { // Register Member

if (totalMembers < maxMembers) {

cout << "Enter Member ID: ";

int id;

if (!(cin >> id)) {

cout << "Invalid Member ID! Please enter a numeric value." << endl;

Beep(500, 500);

cin.clear();

cin.ignore(10000, '\n');

Sleep(1000);

continue;

}



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```
cin.ignore(10000, '\n');
```

```
// Check if member ID already exists
```

```
bool idExists = false;
```

```
for (int i = 0; i < totalMembers; i++) {
```

```
    if (memberID[i] == id) {
```

```
        idExists = true;
```

```
        break;
```

```
    }
```

```
}
```

```
if (idExists) {
```

```
    cout << "Member ID already exists!" << endl;
```

```
    Beep(500, 500);
```

```
    cout << "\nPress Enter to continue...";
```

```
    cin.ignore(10000, '\n');
```

```
    continue;
```

```
}
```

```
memberID[totalMembers] = id;
```

```
// Get member type
```

```
char type;
```

```
while (true) {
```

```
    cout << "Enter Member Type (S for Student, T for
```

```
Teacher): ";
```

```
    cin >> type;
```

```
    cin.ignore(10000, '\n');
```

```
    type = toupper(type);
```

```
    if (type == 'S' || type == 'T') {
```

```
        break;
```

```
    }
```

```
    cout << "Invalid type! Please enter S or T." <<
```

```
endl;
```



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Beep(500, 500);

}

memberType[totalMembers] = type;

// Member name validation (letters and spaces only)

string name;

bool valid = false;

do {

 cout << "Enter Name (letters and spaces only): ";

 getline(cin, name);

 valid = true;

 for (char c : name) {

 if (!((c >= 'a' && c <= 'z') || (c >= 'A' && c

<= 'Z') || c == ' ')) {

 valid = false;

 cout << "Invalid Name! Only letters and

spaces allowed." << endl;

 Beep(500, 500);

 Sleep(500);

 break;

 }

 }

} while (!valid);

memberName[totalMembers] = name;

booksIssuedCount[totalMembers] = 0;

totalMembers++;

cout << "Member registered successfully as "

 << (type == 'S' ? "Student" : "Teacher") << "!" <<

endl;

Beep(1000, 200);

}

else {

 cout << "Member limit reached!" << endl;



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Beep(500, 500);

}

cout << "\nPress Enter to continue...";

cin.ignore(10000, '\n');

}

else if (choice == 5) { // Update Member Details

int id;

cout << "Enter Member ID to update: ";

if (!(cin >> id)) {

cout << "Invalid Member ID! Please enter a numeric

value." << endl;

Beep(500, 500);

cin.clear();

cin.ignore(10000, '\n');

Sleep(1000);

continue;

}

cin.ignore(10000, '\n');

bool found = false;

for (int i = 0; i < totalMembers; i++) {

if (memberID[i] == id) {

found = true;

// Member name validation (letters and spaces only)

string name;

bool valid = false;

do {

cout << "Enter new name (letters and spaces

only): ";

getline(cin, name);

valid = true;

for (char c : name) {

if (!((c >= 'a' && c <= 'z') || (c >= 'A'



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```
&& c <= 'Z') || c == ' ') {
```

```
    valid = false;
```

```
    cout << "Invalid Name! Only letters
```

```
and spaces allowed." << endl;
```

```
    Beep(500, 500);
```

```
    Sleep(500);
```

```
    break;
```

```
    }
```

```
    }
```

```
    } while (!valid);
```

```
    memberName[i] = name;
```

```
    // Update member type
```

```
    char type;
```

```
    while (true) {
```

```
        cout << "Enter new Member Type (S for Student,
```

```
T for Teacher): ";
```

```
        cin >> type;
```

```
        cin.ignore(10000, '\n');
```

```
        type = toupper(type);
```

```
        if (type == 'S' || type == 'T') {
```

```
            break;
```

```
        }
```

```
        cout << "Invalid type! Please enter S or T."
```

```
<< endl;
```

```
        Beep(500, 500);
```

```
    }
```

```
    memberType[i] = type;
```

```
    cout << "Member details updated!" << endl;
```

```
    Beep(1000, 200);
```

```
    break;
```

```
}
```



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```
    }

    if (!found) {
        cout << "Member not found!" << endl;
        Beep(500, 500);
    }
    cout << "\nPress Enter to continue...";
    cin.ignore(10000, '\n');
}

else if (choice == 6) { // Total Books Issued
    int count = 0;
    for (int i = 0; i < totalBooks; i++) {
        if (issuedBooks[i] != 0) count++;
    }
    cout << "Total books issued: " << count << endl;
    cout << "\nPress Enter to continue...";
    cin.ignore(10000, '\n');
}

else if (choice == 8) { // Update Book
    cout << "Enter Book ID to update: ";
    int id;
    if (!(cin >> id)) {
        cout << "Invalid Book ID! Please enter a numeric value."
<< endl;

        Beep(500, 500);
        cin.clear();
        cin.ignore(10000, '\n');
        Sleep(1000);
        continue;
    }
    cin.ignore(10000, '\n');

    bool found = false;
    for (int i = 0; i < totalBooks; i++) {
```



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```
if (bookID[i] == id) {
    found = true;
    string input;
    bool valid = false;

    // Title validation
    do {
        cout << "Enter new Title (letters, numbers,
and spaces only): ";

        getline(cin, input);
        valid = true;
        for (char c : input) {
            if (!((c >= 'a' && c <= 'z') || (c >= 'A'
&& c <= 'Z') ||

(c >= '0' && c <= '9') || c == '

')) {
            valid = false;
            cout << "Invalid Title! Only
letters, numbers and spaces allowed." << endl;
            Beep(500, 500);
            Sleep(500);
            break;
        }
    } while (!valid);
    bookTitle[i] = input;

    // Author validation (letters and spaces only)
    do {
        cout << "Enter new Author (letters and spaces
only): ";

        getline(cin, input);
        valid = true;
        for (char c : input) {
```




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```
        if (!(c >= 'a' && c <= 'z') || (c >= 'A'
&& c <= 'Z') || c == ' ')) {
            valid = false;
            cout << "Invalid Author! Only
letters and spaces allowed." << endl;
            Beep(500, 500);
            Sleep(500);
            break;
        }
    }
} while (!valid);
bookAuthor[i] = input;

// Category validation
do {
    cout << "Enter new Category (letters, numbers,
spaces, and underscores only): ";
    getline(cin, input);
    valid = true;
    for (char c : input) {
        if (!(c >= 'a' && c <= 'z') || (c >= 'A'
&& c <= 'Z') ||
(c >= '0' && c <= '9') || c == ' '
|| c == '_')) {
            valid = false;
            cout << "Invalid Category! Only
letters, numbers, spaces and underscores allowed." << endl;
            Beep(500, 500);
            Sleep(500);
            break;
        }
    }
} while (!valid);
bookCategory[i] = input;
```



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```
// Price validation
float price;
while (true) {
    cout << "Enter new Price: $";
    if (!(cin >> price)) {
        cout << "Invalid Price! Please enter a
numeric value." << endl;

        Beep(500, 500);
        cin.clear();
        cin.ignore(10000, '\n');
        Sleep(500);
    }
    else {
        cin.ignore(10000, '\n');
        break;
    }
}
bookPrice[i] = price;
cout << "Book updated successfully!" << endl;
Beep(1000, 200);
break;
}

}

if (!found) {
    cout << "Book not found!" << endl;
    Beep(500, 500);
}
cout << "\nPress Enter to continue...";
cin.ignore(10000, '\n');
}

else if (choice == 9) { // Remove Book
    cout << "Enter Book ID to remove: ";
```



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```
int id;
if (!(cin >> id)) {
    cout << "Invalid Book ID! Please enter a numeric value."
<< endl;

    Beep(500, 500);
    cin.clear();
    cin.ignore(10000, '\n');
    Sleep(1000);
    continue;
}
cin.ignore(10000, '\n');

bool found = false;
for (int i = 0; i < totalBooks; i++) {
    if (bookID[i] == id) {
        found = true;
        // If book is issued, decrement member's issued
count

        if (issuedBooks[i] != 0) {
            int memberId = issuedBooks[i];
            for (int j = 0; j < totalMembers; j++) {
                if (memberID[j] == memberId) {
                    booksIssuedCount[j]--;
                    break;
                }
            }
        }

        // Remove the book
        for (int j = i; j < totalBooks - 1; j++) {
            bookID[j] = bookID[j + 1];
            bookTitle[j] = bookTitle[j + 1];
            bookAuthor[j] = bookAuthor[j + 1];
            bookCategory[j] = bookCategory[j + 1];
```



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```
        bookPrice[j] = bookPrice[j + 1];

        issuedBooks[j] = issuedBooks[j + 1];

    }

    totalBooks--;

    cout << "Book removed successfully!" << endl;

    Beep(1000, 200);

    break;

    }

}

if (!found) {

    cout << "Book not found!" << endl;

    Beep(500, 500);

}

cout << "\nPress Enter to continue...";

cin.ignore(10000, '\n');

}

else if (choice == 10) { // View All Books

    cout << "\n===== ALL BOOKS =====" << endl;

    for (int i = 0; i < totalBooks; i++) {

        cout << "ID: " << bookID[i] << " | Title: " <<

bookTitle[i]

        << " | Author: " << bookAuthor[i] << " | Category: "

        << bookCategory[i] << " | Price: $" << bookPrice[i]

        << " | Status: " << (issuedBooks[i] != 0 ? "Issued"

: "Available");

        if (issuedBooks[i] != 0) {

            // Find member name

            string issuerName = "Unknown";

            for (int j = 0; j < totalMembers; j++) {

                if (memberID[j] == issuedBooks[i]) {

                    issuerName = memberName[j];

                    break;

                }

            }

        }

    }

}
```



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```
}
    cout << " (Issued to: " << issuerName << ")";
}
    cout << endl;
}
    cout << "\nPress Enter to continue...";
    cin.ignore(10000, '\n');
}
else if (choice == 11) { // Search by Title
    string title;
    cout << "Enter book title to search: ";
    getline(cin, title);
    bool found = false;
    cout << "\n===== SEARCH RESULTS =====" << endl;
    for (int i = 0; i < totalBooks; i++) {
        if (bookTitle[i] == title) {
            found = true;
            cout << "ID: " << bookID[i] << " | Title: " <<
bookTitle[i]
<< " | Author: " << bookAuthor[i] << " |
Category: "
<< bookCategory[i] << " | Price: $" <<
bookPrice[i]
<< " | Status: " << (issuedBooks[i] != 0 ?
"Issued" : "Available");
            if (issuedBooks[i] != 0) {
                // Find member name
                string issuerName = "Unknown";
                for (int j = 0; j < totalMembers; j++) {
                    if (memberID[j] == issuedBooks[i]) {
                        issuerName = memberName[j];
                        break;
                    }
                }
            }
        }
    }
}
```



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```
        cout << " (Issued to: " << issuerName << ")";
    }
    cout << endl;
}

}

if (!found) {
    cout << "No books found with this title!" << endl;
    Beep(500, 500);
}

cout << "\nPress Enter to continue...";
cin.ignore(10000, '\n');
}

else if (choice == 12) { // Search by Author
    string author;
    cout << "Enter author name to search: ";
    getline(cin, author);
    bool found = false;
    cout << "\n===== SEARCH RESULTS =====" << endl;
    for (int i = 0; i < totalBooks; i++) {
        if (bookAuthor[i] == author) {
            if (!found) {
                cout << "Author: " << author << endl;
                found = true;
            }
            cout << "ID: " << bookID[i] << " | Title: " <<
bookTitle[i]
<< " | Category: " << bookCategory[i] << " |
Price: $" << bookPrice[i]
<< " | Status: " << (issuedBooks[i] != 0 ?
"Issued" : "Available");
            if (issuedBooks[i] != 0) {
                // Find member name
                string issuerName = "Unknown";
                for (int j = 0; j < totalMembers; j++) {
```



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```
        if (memberID[j] == issuedBooks[i]) {
            issuerName = memberName[j];
            break;
        }
    }
    cout << " (Issued to: " << issuerName << ")";
}
cout << endl;
}
}
if (!found) {
    cout << "No books found by this author!" << endl;
    Beep(500, 500);
}
cout << "\nPress Enter to continue...";
cin.ignore(10000, '\n');
}
else if (choice == 13) { // Search by Category
    string category;
    cout << "Enter category to search: ";
    getline(cin, category);
    bool found = false;
    cout << "\n===== SEARCH RESULTS =====" << endl;
    for (int i = 0; i < totalBooks; i++) {
        if (bookCategory[i] == category) {
            found = true;
            cout << "ID: " << bookID[i] << " | Title: " <<
bookTitle[i]
<< " | Author: " << bookAuthor[i] << " |
Category: "
<< bookCategory[i] << " | Price: $" <<
bookPrice[i]
<< " | Status: " << (issuedBooks[i] != 0 ?
"Issued" : "Available");
```



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```
if (issuedBooks[i] != 0) {
    // Find member name
    string issuerName = "Unknown";
    for (int j = 0; j < totalMembers; j++) {
        if (memberID[j] == issuedBooks[i]) {
            issuerName = memberName[j];
            break;
        }
    }
    cout << " (Issued to: " << issuerName << ")";
}
cout << endl;
}
}
if (!found) {
    cout << "No books found in this category!" << endl;
    Beep(500, 500);
}
cout << "\nPress Enter to continue...";
cin.ignore(10000, '\n');
}
else if (choice == 14) { // Exit
    cout << "Are you sure you want to exit? (y/n): ";
    char confirm;
    cin >> confirm;
    cin.ignore(10000, '\n');
    if (confirm == 'y' || confirm == 'Y') {
        cout << "Exiting the program..." << endl;
        Beep(1000, 500);
        break;
    }
}
}
return 0;
```




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}

Menu :

```
C:\Users\Lenovo\source\repos\Project2\Debug\Project2.exe

===== LIBRARY MANAGEMENT SYSTEM =====

1. Issue Book
2. Return Book
3. Buy Book
4. Register Member
5. Update Member Details
6. Total Books Issued
7. Add Book
8. Update Book
9. Remove Book
10. View All Books
11. Search Book by Title
12. Search Book by Author
13. Search Book by Category
14. Exit

Enter your choice (1-14):
```

Sub Menu:

```
C:\Users\Lenovo\source\repos\Project2\Debug\Project2.exe

===== LIBRARY MANAGEMENT SYSTEM =====

1. Issue Book
2. Return Book
3. Buy Book
4. Register Member
5. Update Member Details
6. Total Books Issued
7. Add Book
8. Update Book
9. Remove Book
10. View All Books
11. Search Book by Title
12. Search Book by Author
13. Search Book by Category
14. Exit

Enter your choice (1-14): 4
Enter Member ID: 72237635
Enter Member Type (S for Student, T for Teacher): s
Enter Name (letters and spaces only): Maira Inshal Zainab Aniq
Member registered successfully as Student!

Press Enter to continue...
```

Flow Chart :



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